

Bioassay Suite — visual language

RStudio/Jupyter typographic feel · Okabe-Ito colorblind-safe palette · IBM Plex type system. Quiet, dense, trustworthy.

Typography		IBM Plex Sans · Mono · Serif
28 / 500	Bioassay Analysis Report	Plex Sans · Display
22 / 500	Step 2 — Plate Layout	Plex Sans · H1
17 / 600	Standard concentrations	Plex Sans · H2
14 / 500	Section heading	Plex Sans · H3
13 / 400	Body — the app auto-detects plate regions in .xlsx, .csv, and .txt files.	Plex Sans · Body
12 / 400	Help text — values greater than 1 are interpreted as pre-concentration.	Plex Sans · Small
12 / 400 mono	R ² = 0.9919 · LLOQ 62.8 pg/mL · n = 8	Plex Mono · Numbers

Color · Okabe-Ito (colorblind-safe)

8-color palette + neutrals

Black

#000000

Orange

#e69f00

Sky blue

#56b4e9

Bluish green

#009e73

Yellow

#f0e442

Blue

#0072b2

Vermillion

#d55e00

Reddish purple

#cc79a7

BRAND & NEUTRALS

Accent (navy)

#1f3a5f

Accent soft

#e8edf4

Ink

#0f1722

Ink-2

#2a3340

Ink-3

#5b6573

Surface-2

#f1f3f5

Line

#e3e6ea

BG

#f7f8f9

SEMANTIC — MAPPED TO OKABE-ITO SO THEY REMAIN COLORBLIND-SAFE

● PASS · #009e73

● WARN · #e69f00

● FAIL · #d55e00

● INFO · #0072b2

Generate report →

Save layout

Reset

Small

⌘ S

STEP 01 / 05

Configuration

Choose an assay type and define standard concentrations. Or load a preset to skip ahead.

Next: Plate Layout →

Quick start

3 presets · or configure manually below

RBA · Saxitoxin

8 standards, triplicate · demo data included

→

ELISA · Cortisol

Cayman kit · 8-point duplicate · demo data

→

ELISA · Custom

Blank template — manual configuration

→

Presets fill the assay type, plate layout, and standard concentrations. You can still edit anything.

A

Assay

Pick the assay family. Affects which control wells (Blank, NSB, B0) are required and how %B/B0 is computed.

Type of assay

ELISA — Enzyme-Linked Immunosorbent Assay

ELISA measures analyte concentration via antibody-enzyme reactions.

Analyte

Cortisol

Concentration units

pg/mL

B

Standards

Enter the nominal concentration of each standard. Highest first is conventional.

Number of standards

8

Enter concentrations in pg/mL, e.g. 4000, 1600, 640...

S1 (pg/mL)

4000

S2 (pg/mL)

1600

S3 (pg/mL)

640

S4 (pg/mL)

256

S5 (pg/mL)

102.4

S6 (pg/mL)

41

S7 (pg/mL)

16.4

S8 (pg/mL)

6.6

STEP 02 / 05

Plate layout

Define what each well contains. Four layers — switch between them on the left. All four edits stay synchronized.

📁 Save layout

Next: Upload →

Load preset

RBA · STX 8 standards (triplicate) ▾

Import layout (CSV / Excel)

📁 Browse...

↶ Undo

↷ Redo

Layers

Sample Type

01 Standard, Sample, Blank, NSB, 5 types

B0

Sample ID

02 Identifier per well

40 IDs

Dilution Fraction

03 Strength remaining (1 = undiluted)

8 levels

Replicate Groups

04 Wells averaged together

14 groups

BULK ACTION

1

Set all wells

Sample Type

Reset

● 96 wells

Standard, Sample, Blank, NSB, B0

	1	2	3	4	5	6	7	8	9	10	11	12
A	Standard	Standard	Standard	Blank	Sample	Sample	Sample	Sample	Sample	Sample	Sample	Sample
B	Standard	Standard	Standard	Blank	Sample	Sample	Sample	Sample	Sample	Sample	Sample	Sample
C	Standard	Standard	Standard	NSB	Sample	Sample	Sample	Sample	Sample	Sample	Sample	Sample
D	Standard	Standard	Standard	NSB	Sample	Sample	Sample	Sample	Sample	Sample	Sample	Sample
E	Standard	Standard	Standard	B0	Sample	Sample	Sample	Sample	Sample	Sample	Sample	Sample
F	Standard	Standard	Standard	B0	Sample	Sample	Sample	Sample	Sample	Sample	Sample	Sample
G	Standard	Standard	Standard	B0	Sample	Sample	Sample	Sample	Sample	Sample	Sample	Sample
H	Standard	Standard	Standard	B0	Sample	Sample	Sample	Sample	Sample	Sample	Sample	Sample

■ Standard

■ Sample

■ Blank

■ NSB

■ B0

STEP 03 / 05

Upload & preview

Drop the plate-reader export. The app auto-detects the 8×12 region and previews it as a heatmap so you can confirm before analysis.

Next: Analysis Settings →

Plate reader file

.xlsx · .csv · .txt

☒ Classic import

☐ Visual plate selector

📄

260220_Cortisol.xlsx

284 KB · uploaded 2 min ago

OK

Replace

Download example

Show plate template

Plate heatmap

auto-detected · 96 / 96 wells

	1	2	3	4	5	6	7	8	9	10	11	12
A	1.63	1.64	1.63	0.05	0.22	0.24	0.19	0.13	0.13	0.18	0.23	0.23
B	1.46	1.45	1.43	0.05	0.28	0.26	0.20	0.16	0.19	0.25	0.28	0.25
C	1.27	1.25	1.23	0.15	0.32	0.28	0.22	0.20	0.25	0.31	0.31	0.26
D	1.07	1.05	1.03	0.15	0.35	0.29	0.24	0.26	0.32	0.36	0.33	0.27
E	0.87	0.85	0.84	1.70	0.36	0.30	0.28	0.33	0.39	0.39	0.34	0.29
F	0.67	0.66	0.67	1.70	0.37	0.32	0.34	0.40	0.44	0.42	0.35	0.32
G	0.48	0.49	0.52	1.70	0.38	0.36	0.40	0.46	0.48	0.43	0.37	0.37
H	0.31	0.34	0.37	1.70	0.41	0.41	0.47	0.52	0.50	0.44	0.40	0.43

0.05

2.00

absorbance · perceptually uniform · CB-safe

Import summary

Format	labeled_with_headers
Wells detected	96 / 96
Partial plate	No
Decimal locale	period (.)
Overflow markers	0 found
Wavelength	single (450 nm)

✓ Plate region detected automatically. Heatmap matches expected dilution gradient (high→low across rows on standards).

Raw data sample

first 3 rows

	1	2	3	4	5	6	7	8
A	0.16	0.21	0.17	0.19	0.18	0.19	0.16	0.10
B	0.16	0.31	0.30	0.32	0.32	0.32	0.56	0.51
C	0.16	0.44	0.45	0.17	0.17	0.18	0.23	0.23

Analysis settings

Defaults are conservative and match common immunoassay protocols. Change them only if your assay or kit guidance specifies otherwise.

Next: Generate Report →

Curve fitting & QC

4-parameter logistic · drc package

DRC regression weighting

How to weight residuals during 4PL curve fitting. Multiple selections fit several curves and compare side-by-side.

☒ Unweighted

Equal weight on all standards

☐ 1/Y (moderate)

Down-weights high-response standards

☐ 1/Y² (recommended for immunoassays)

Standard immunoassay practice — reflects proportional error

☐ Auto (data-driven)

Picks weighting via Brown-Forsythe heteroscedasticity test

Quantification range (%B/B0)

Samples whose response falls outside this band are flagged <LLOQ or >ULOQ.

Lower bound

Upper bound

20

80

Conventional bounds: 20%–80% . Tighter bounds (e.g. 30–70) give more conservative LLOQ/ULOQ.

Confidence interval method

Used for replicate-group concentration estimates and the report's CI column.

☒ t-distribution (default)

Fast · assumes normal residuals

☐ Bootstrap (1000 resamples)

Slower · no normality assumption

Outlier detection

Optional — runs per replicate group. Disabled by default; enable for assays with noisy replicates.

☐ Enable outlier detection

Min replicates for test

3

☒ Assume normality

Dixon's Q (n=3-5) · Grubbs (n≥6)

☐ Test with Shapiro-Wilk

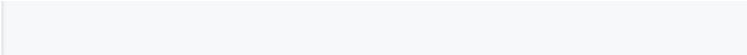
Falls back to MAD-based test if non-normal



Maximum CV for standards

30

%



STEP 05 / 05

Generate report

Pick output formats and language, then generate. The pre-flight panel below confirms your data is ready.

Output

all formats use the same analysis

Report formats

☒ HTML

interactive Plotly · default

☐ DOCX

Word · static figures

☐ PDF

requires LaTeX engine

Languages

☒ English

☐ Español

☒ Generate compact report (no exclusion audit appendix)

⬇️

Generate report

Pre-flight check

6 / 6 passed

✓

Plate data uploaded

96 / 96 wells

✓

Standard wells defined

24 wells

✓

Dilution factors valid

all numeric

Run notes

optional · appears in report

Cortisol ELISA · Cayman kit lot 0628942

Tissue: zebrafish whole-body homogenate, n=12

Notes: B1/B2 wells appeared turbid prior to read — flagged for review.

Last run

Generated	2026-04-22 14:08
Output	analysis_report.html
Folder	~/Documents/2026-04-22_03

Analysis report · run 03 of 2026-04-22

Bioassay Analysis Report

Assay ELISA — cortisol Date 2026-04-22 Analyst moellerkr Software Bioassay Suite v0.9.0

R² · CURVE FIT

0.9919

LL.4 · 1/Y² weighted

RMSE

4.66

%B/B0

LL0Q

62.8

pg/mL

ULOQ

1,387

pg/mL

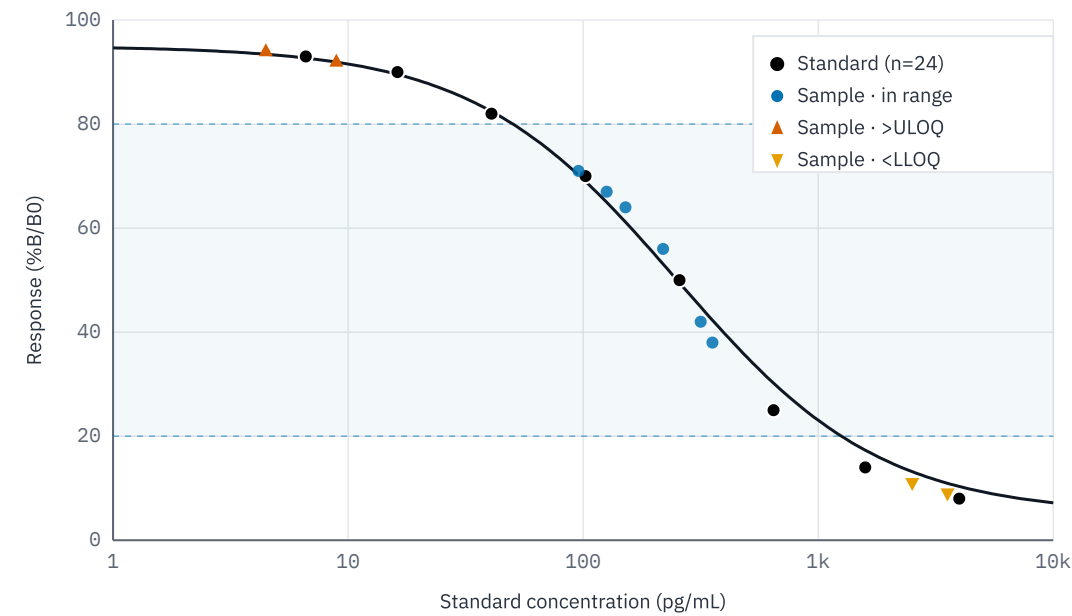
OVERALL

●

 Pass

all 6 checks ok

Executive summary		at-a-glance run status	
Assay	ELISA — cortisol	Model	4PL · LL.4 · auto regression
Standards	8 levels · 24 wells used	Quant. range	20% – 80% B/B0
Sample wells	72	Controls	Blank 0.156 · NSB 0.159 · B0 0.966 · ● valid hierarchy
Replicate groups	14	CI method	t-distribution (95%)



Shaded band shows the validated quantification range (20–80% B/B0). Triangles flag samples above ULOQ (orange-vermillion) or below LLOQ (orange).

§ 02

Sample results

GROUP	SAMPLE IDS	TYPE	N	MEAN (PG/ML)	SD	CV %	%B/B0	RANGE	STATUS
AA	A4,A5	Sample	2	8,649.1	945.92	8.21	3.89	● >ULOQ	● Extrapolated
AB	C4,C5	Sample	2	1,116.9	420.10	8.76	11.92	● >ULOQ	● Interpolated
AC	E4,E5	Sample	2	1,898	283.44	14.23	13.06	● >ULOQ	● Interpolated
AD	G4,G5	Sample	2	4,574	371.99	8.13	3.28	● >ULOQ	● Extrapolated
AE	B6,B5	Sample	2	1,366.7	44.71	3.22	20.51	● in range	● Interpolated
AF	D4,D5	Sample	2	3,472	105.59	3.05	7.42	● >ULOQ	● Interpolated
AG	F4,F5	Sample	2	510.4	5.05	0.99	43.93	● in range	● Interpolated
AH	H4,H5	Sample	2	1,395.3	19.38	1.39	19.07	● in range	● Interpolated
AI	I4,I5	Sample	2	8,015.5	2099.22	79.01	10.04	● >ULOQ	● Extrapolated